Q09 Deeper Project Suggestions

1. **Dedup aggressiveness sweep:** implement exact, MinHash, and suffix-array-style substring dedup over a small public corpus (e.g., 100MB of C4 or The Stack samples) at varying Jaccard thresholds and band sizes; train a small LM at each setting and plot the frontier of (unique tokens retained) vs. (loss and verbatim-memorization rate on held-out prompts). Locate the FineWeb-style point where more dedup starts to hurt.
2. **Mixture-weight scaling study:** pick 4–5 domains (web, code, math, books) and fit a small-scale analogue of DoReMi or data-mixing laws: train tiny models (10–50M params) on a grid of mixtures, fit a predictive function of validation loss vs. mixture, and test whether the predicted optimum transfers to a model 4–10× larger.
3. **Repetition vs. reweighting:** run a controlled 2×2 — {repeat whole corpus, upsample one domain} × {1 epoch, 4 epochs} — at fixed token budget and measure both in-domain and out-of-domain loss, to test whether the 4-epoch rule survives under targeted up-sampling.
4. **Tokenizer-fertility audit:** train BPE tokenizers of the same vocabulary size on different mixtures (web-only vs. web+code+math), tokenize a fixed evaluation suite with each, and quantify how declared domain weights shift when accounting is done in tokens, bytes, and unique-word types; report where the discrepancy is largest.